

MA 102 (Mathematics II - Ordinary Differential Equations)

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Solutions - Tutorial Sheet No. 2
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Date of Discussion:

Lectures 5-6: Existence and uniqueness of solutions, applications of first-order ODEs.

1. Discuss the existence and uniqueness of the solution of the following initial value problems.

(i) $\frac{dy}{dx} = y^{1/3}, y(0) = 0.$

(ii) $\frac{dy}{dx} = y^{1/3}, y(0) = 1.$

(iii) $\frac{dy}{dx} = x^2 + y^2, y(0) = 0.$

(iv) $\frac{dy}{dx} = \sqrt{1 - y^2}, y(\pi/2) = 1.$

(v) $\ln\left(\frac{dy}{dx}\right) = y^2, y(x_0) = y_0, \text{ where } x_0, y_0 \in \mathbb{R}.$

Solution:

(i) Note that $f(x, y) = y^{1/3}$ is continuous in the rectangle $R = \{(x, y) : |x| \leq a, |y| \leq b\}$ for any $a, b > 0$.

Therefore, by Peano's theorem, the IVP admits at least one solution. However, note that $f(x, y) = y^{1/3}$ is not Lipschitz continuous in R . This is because

$$\frac{|f(x, y) - f(x, 0)|}{|y|} = \frac{|y^{1/3}|}{|y|} = |y^{-2/3}| \rightarrow \infty \text{ as } y \rightarrow 0.$$

It can be found that $y \equiv 0, y_c(x) = \begin{cases} 0, & x \leq c, \\ (\frac{2}{3}(x - c))^{3/2}, & c < x, \end{cases}$ for $c > 0$ are infinite number of solutions of the IVP.

(ii) For this IVP, $f(x, y) = y^{1/3}$ is continuous in the rectangle $R = \{(x, y) : |x| \leq a, |y - 1| \leq b\}$ for any $a, b > 0$. It is further Lipschitz continuous. Therefore, this IVP admits a unique solution in $[-h, h]$ for some $h > 0$.

(iii) For this IVP, $f(x, y) = x^2 + y^2$ is continuous in the rectangle $R = \{(x, y) : |x| \leq a, |y| \leq b\}$ for any $a, b > 0$. It is further Lipschitz continuous. Therefore, this IVP admits a unique solution in $[-h, h]$ for some $h > 0$.

(iv) For this IVP, $f(x, y) = \sqrt{1 - y^2}$ is continuous in the rectangle $R = \{(x, y) : |x - \frac{\pi}{2}| \leq a, |y - 1| \leq b\}$ for any $a, b > 0$. Therefore, by Peano's theorem, the IVP

admits at least one solution. However, it is not Lipschitz continuous. It can be found that $y(x) \equiv 1$ and $y(x) = \sin x$ are two solutions of the IVP.

(v) The ODE can be written as $y' = e^{y^2}$. This IVP has a unique solution.

2. Find the functions y_1, y_2 , and y_3 using the Method of Successive Approximations for the differential equation $\frac{dy}{dx} = y - x + 1$, with the initial condition $y(0) = 0$. Then find the exact solution to this initial value problem by taking the limit of the sequence of approximations.

Solution:

Here, $x_0 = 0$, $y_0 = 0$ and the initial approximation is $y_0(x) = y_0 = 0$. The first approximation is given by

$$y_1(x) = \int_0^x f(\tau, y_0(\tau)) d\tau = \int_0^x (1 - \tau) d\tau = x - \frac{x^2}{2}.$$

The second approximation is given by

$$y_2(x) = \int_0^x f(\tau, y_1(\tau)) d\tau = \int_0^x (1 - \tau + \tau - \frac{\tau^2}{2}) d\tau = x - \frac{x^3}{2 \cdot 3}.$$

The third approximation is given by

$$y_3(x) = \int_0^x f(\tau, y_2(\tau)) d\tau = \int_0^x (1 - \tau + \tau - \frac{\tau^3}{3!}) d\tau = x - \frac{x^4}{4!}.$$

The n -th approximation is

$$y_n(x) = x - \frac{x^{n+1}}{(n+1)!} \rightarrow x \text{ as } n \rightarrow \infty.$$

Therefore $y(x) = x$ is the unique solution of the IVP.

3. Find p such that the IVP $\frac{dy}{dx} = y^p$, $y(0) = 0$ admits a unique solution.

Solution:

The IVP is given as $\frac{dy}{dx} = y^p$, $y(0) = 0$. Consider the rectangle $R = \{(x, y) : |x| \leq a, |y| \leq b\}$ for any $a, b > 0$. For $(f(x, y) = x^p)$ to be continuous in R , we must have $p \geq 0$.

Now, for Lipschitz continuity, we must have

$$\left| \frac{d}{dy}(y^p) \right| \leq m \text{ in } R \text{ for some } m > 0.$$

That is, we must have $|py^{p-1}| \leq m$ in R .

This is possible only when $p \geq 1$. Therefore, only for $p \geq 1$, the given IVP admits a unique solution.

4. Show that the solution $y(x)$ of the initial value problem $\frac{dy}{dx} = 2 + \sin y$, $y(0) = 0$ cannot vanish for $x > 0$.

Solution: Given that $\frac{dy}{dx} = 2 + \sin y$, $y(0) = 0$. We know that $|\sin y| \leq 1 \forall y \in \mathbb{R}$.

From the given equation, it is clear that $y' > 0$. This means that the function is strictly increasing for $x > 0$.

Now, $y(0) = 0 \Rightarrow y(x) > 0 \forall x > 0$. This implies that the solution of the given IVP cannot vanish for $x > 0$.

5. Consider the IVP:

$$\frac{dy}{dx} = f(x, y), \quad y(x_0) = y_0.$$

Let $f(x + T, y) = f(x, y)$ and $y(x_0 + T) = y_0$, where $f(x, y)$ satisfies the assumptions of Picard's theorem. Then prove that the solution y is also periodic of period T , i.e. $y(x + T) = y(x)$ for all x .

Solution:

It is given that f satisfies the assumptions of Picard's theorem and $f(x + T, y) = f(x, y)$, $y(x_0 + T) = y_0$. We claim that $y(x + T) = y(x) \quad \forall x$.

Now, set $z(x) = y(x + T)$. It implies that

$$\begin{aligned} \frac{d}{dz}(z(x)) &= \frac{d}{dx}(y(x + T)) \\ &= f(x + T, y(x + T)) \quad \text{since } y \text{ is a solution of } y' = f(x, y) \\ &= f(x, z), \end{aligned}$$

with $z(x_0) = y(x_0 + T)$.

Therefore, $z(x)$ and $y(x)$ satisfy the IVP. Since this IVP has a unique solution (f satisfies the assumptions of Picard's theorem), we must have

$$\begin{aligned} z(x) &\equiv y(x) \quad \forall x \\ \Rightarrow y(x + T) &= y(x) \quad \forall x \\ \Rightarrow y &\text{ is periodic.} \end{aligned}$$

6. For the initial value problem

$$\frac{dy}{dx} = x + e^x \sin(xy), \quad y(0) = y_0 = 0,$$

estimate the variation of the solution in the interval $[0, 1]$ if y_0 is perturbed by 0.01.

Solution:

Result: Let f be continuous and satisfy a Lipschitz condition in y , with Lipschitz constant k , in a domain D of the xy -plane; and let (x_0, y_0) be a fixed point of D . Assume there exist $\delta > 0$ and $h > 0$ such that for each Y_0 satisfying $|Y_0 - y_0| \leq \delta$, the IVP

$$\frac{dy}{dx} = f(x, y), \quad y(x_0) = Y_0$$

possesses a unique solution $\phi(x, Y_0)$ defined and contained in D on $|x - x_0| \leq h$.

If ϕ denotes the unique solution of the equation when $Y_0 = y_0$, and $\bar{\phi}$ denotes the unique solution when $Y_0 = \bar{y}_0$, where $|\bar{y}_0 - y_0| = \delta_1 \leq \delta$, then

$$|\bar{\phi} - \phi(x)| \leq \delta_1 e^{Lh} \quad \text{on } |x - x_0| \leq h.$$

For the given problem, we have

$$|\bar{\phi} - \phi(x)| \leq 0.01e^L \quad \text{since we are given } h = 1.$$

Here, $L = \sup_R |xe^x \cos xy| \leq e^1 = e$, where

$R = \{(x, y) : |x| \leq 1, |y| \leq b\}$. This implies that

$$|\bar{\phi} - \phi(x)| \leq 0.01e^e.$$

7. Let $y(x, \lambda)$ denote the unique solution of the initial value problem

$$(IVP)_\lambda : \quad \frac{dy}{dx} = \lambda + \cos y, \quad y(0) = y_0 = 0.$$

Then obtain an upper estimate for $|y(x, \lambda_1) - y(x, \lambda_2)|$ in the interval $[0, 1]$.

Solution:

We have

$$|\phi(x) - \bar{\phi}(x)| \leq e^{Lh} \max |f(x, y) - \bar{f}(x, y)|.$$

Now,

$$\begin{aligned} f(x, y) &= \lambda_1 + \cos y, \\ \bar{f}(x, y) &= \lambda_2 + \cos y \end{aligned}$$

implies

$$|f(x, y) - \bar{f}(x, y)| = |\lambda_1 - \lambda_2|.$$

Further, $L = \sup |\sin y| = 1$. Therefore,

$$|\phi(x) - \bar{\phi}(x)| \leq e|\lambda_1 - \lambda_2|.$$

8. Show that the orthogonal trajectories of a given family of curves $f(x, y) = c$ are represented by the differential equation

$$\frac{dy}{dx} = \frac{\partial f}{\partial y} \bigg/ \frac{\partial f}{\partial x}.$$

Solution:

The family of curves is given by $f(x, y) = c$. Differentiating this, we get $df = 0$ which in turn gives us

$$\frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy = 0$$

giving the differential equation for the given family as $\frac{dy}{dx} = -\frac{\partial f}{\partial x} \bigg/ \frac{\partial f}{\partial y}$.

Hence, the differential equation for the orthogonal trajectories will be given by

$$\frac{dy}{dx} = \frac{\partial f}{\partial y} \bigg/ \frac{\partial f}{\partial x}.$$

9. Show that the family of parabolas $y^2 = 2cx + c^2$ where $c \in \mathbb{R}$ is an arbitrary constant is self-orthogonal. (A given family of curves is said to be self-orthogonal if its family of the orthogonal trajectories is the same as the given family.)

Solution:

Differentiating $y^2 = 2cx + c^2$, we get $c = yy'$ so that the given equation can be written as

$$y^2 = 2yy'x + (yy')^2.$$

The differential equation for the orthogonal trajectories will be given by

$$y^2 = -\frac{2yx}{y'} + \frac{y^2}{y'^2},$$

which gives us

$$y^2 = 2xyy' + (yy')^2,$$

which is the same as the differential equation for the given family. Hence, the given family is self-orthogonal.

10. Find the value of n such that the curves $x^n + y^n = c_1$ are the orthogonal trajectories of the family $y = \frac{x}{1 - c_2x}$, where c_1 and c_2 are arbitrary constants.

Solution:

Differentiating $x^n + y^n = c_1$, we get its differential equation as $y' = -x^{n-1}y^{1-n}$. Therefore, the differential equation of its orthogonal trajectories will be given by $y' = x^{1-n}y^{n-1}$. But it is given that the orthogonal trajectories are given by $y = \frac{x}{1 - c_2x}$.

Differentiating it, we get $y' = \frac{1}{(1 - c_2x)^2}$. Eliminating c_2 , we have $y' = x^{-2}y^2$. Comparing it with $y' = x^{1-n}y^{n-1}$, both of which are the same, we get $n = 3$.

11. An object cools in air of constant temperature 20°C . Ten minutes after the object began to cool, its temperature was observed to be 75°C , and ten minutes later, its temperature was observed to be 50°C . What was its temperature when it began to cool?

Solution:

The differential equation can be written as

$$\frac{dT}{dt} = -k(T - 20).$$

Its solution is given by

$$T(t) = 20 + ce^{-kt}.$$

using the condition $T(10) = 75$, we get $c = 55e^{10k}$. Using the condition $T(20) = 50$, we get $e^{10k} = \frac{11}{6}$. Therefore we can write the solution as

$$T(t) = 20 + 55e^{k(10-t)}.$$

When the object started to cool at $t = 0$, we have the temperature

$$T(0) = 20 + 55e^{10k} = 20 + 55 \times \frac{11}{6} \approx 120.8.$$

12. Follow the current flow in the RL circuit:

$$L\frac{dI}{dt} + RI = E(t).$$

Find current $I(t)$ when the initial current is given as I_0 , and the emf imposed at time $t = 0$ is given by

(a) $E = E_0e^{-kt}$, $k > 0$,

(b) $E = E_0 \sin \omega t$.

Solution:

We write the differential equation in the form

$$\frac{dI}{dt} + \frac{R}{L}I = \frac{E(t)}{L}.$$

The integrating factor is $e^{Rt/L}$ and hence the solution can be written as

$$I(t)e^{Rt/L} = \int \frac{E(t)}{L} e^{Rt/L} dt + A.$$

(a) Here. $E(t) = E_0 e^{-kt}$, $k > 0$. The solution can be found as

$$I(t) = \frac{E_0}{R - kL} e^{-kt} + A e^{-Rt/L}.$$

Use the initial condition $I(0) = I_0$ to get $A = I_0 - \frac{E_0}{R - kL}$ so as to write the solution as

$$I(t) = \frac{E_0}{R - kL} (e^{-kt} - e^{-Rt/L}) + I_0 e^{-Rt/L}.$$

(b) Here, $E(t) = E_0 \sin \omega t$. The solution takes the form

$$I(t) e^{Rt/L} = \frac{E_0}{L} \int \sin \omega t e^{Rt/L} dt + B.$$

Recall the integration formula

$$\int e^{ax} \sin bx \, dx = \frac{e^{ax}}{a^2 + b^2} (a \sin bx - b \cos bx).$$

Using the above, the solution can be obtained as

$$I(t) = \frac{E_0 L}{R^2 + L^2 \omega^2} \left(\frac{R}{L} \sin \omega t - \omega \cos \omega t \right) + B e^{-Rt/L}.$$

Use the initial condition $I(0) = I_0$ to get $B = I_0 + \frac{E_0 L \omega}{R^2 + \omega^2 L^2}$. Hence, the solution can be found as

$$I(t) = \frac{E_0 L}{R^2 + L^2 \omega^2} \left(\frac{R}{L} \sin \omega t - \omega \cos \omega t \right) + \left(I_0 + \frac{E_0 L \omega}{R^2 + \omega^2 L^2} \right) e^{-Rt/L}.$$

The End
